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start_state = [False, False, False, False]
goal_state = [True, True, True, True]
Queue = [start_state]
visited = [start_state]
elements = ['Man', 'Lion', 'Goat', 'Grass']
starting_side = []
target_side = []
shortest_path = [[start_state]]

def ValidationFunction(current_state):
    if current_state[1] == current_state[2] and current_state[0] != current_state[1]:
        return False
    if current_state[2] == current_state[3] and current_state[0] != current_state[2]:
        return False
    else:
        return True

def GenerationFunction(current_state):
    generated_states = []
    for index in range(0, len(current_state)):
        if index == 0:
            temp_state = current_state.copy()
            temp_state[0] = not temp_state[0]
            generated_states.append(temp_state)
        else:
            if current_state[index] == current_state[0]:
                temp_state = current_state.copy()
                temp_state[0] = not temp_state[0]
                temp_state[index] = not temp_state[index]
                generated_states.append(temp_state)
    return generated_states

while Queue:
    if len(shortest_path) == 1:
        print("starting side: ['Man', 'Lion', 'Goat', 'Grass']")
        print("target side: []")
        print('')
    starting_side.clear()
    target_side.clear()

    current_state = Queue.pop(-1)
    if current_state == goal_state:
        break
    generated_states = GenerationFunction(current_state)
    for state in generated_states:

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if state not in visited :
    validated = ValidationFunction(state)
    if validated == True:
        visited.append(state)
        if len(shortest_path) > 1:
            if state[0] != shortest_path[-1][0]:
                shortest_path.append(state)
                for index in range(len(shortest_path[-1])):
                    if shortest_path[-1][index] == False:
                        starting_side.append(elements[index])
                    elif shortest_path[-1][index] == True:
                        target_side.append(elements[index])
                print('starting side: ', starting_side)
                print('target side: ', target_side)
                print('')
            elif len(shortest_path) <= 1:
                shortest_path.append(state)
                for index in range(len(shortest_path[-1])):
                    if shortest_path[-1][index] == False:
                        starting_side.append(elements[index])
                    elif shortest_path[-1][index] == True:
                        target_side.append(elements[index])
                print('starting side: ', starting_side)
                print('target side: ', target_side)
                print('')
        Queue.append(state)

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